

Gabriel Mukobi

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Summary:

AI security researcher specializing in assessing and mitigating the security risks of advanced AI systems, with experience across AI/ML research, policy analysis, and software engineering in both small-team and large bureaucratic environments.

Experience:

AI Security Researcher, U.S. Center for AI Standards and Innovation - Jul 2024-Present - 40 h/wk - Washington, DC - [nist.gov/caisi](https://www.nist.gov/caisi)

Conducted frontier AI research to protect national security, public safety, and economic competitiveness from AI-related threats. Focused on both improving the security of AI systems and securing the U.S. AI ecosystem against adversaries. Built evaluations, delivered briefings and technical analyses, and coordinated with Government and industry partners.

Research Fellow, RAND - Apr 2024-Jul 2024 - 40 h/wk - Washington, DC - [rand.org](https://www.rand.org)

Informed and improved AI policy decision-making through technical and policy research and analysis. Conducted rapid technical analyses about emerging AI issues and wrote policy memos for key government stakeholders.

Technical AI Research Fellow, ERA Fellowship - Jul 2023-Sep 2023 - 40 h/wk - Cambridge, UK - erafellowship.org

Led self-directed research building evaluations of the cooperative AI capabilities of language models in multi-agent environments.

Gameplay Engineering Intern, Respawn Entertainment - Jun 2022-Sep 2022 - 40 h/wk - Remote - [ea.com](https://www.ea.com)

Engineered core gameplay and AI features on Respawn's unreleased Star Wars bounty hunter first-person shooter.

Gameplay Engineering Intern, Riot Games - Jun 2021-Sep 2021 - 40 h/wk - Remote - riotgames.com

Designed and implemented core features as a Software Engineering Intern on the gameplay team of [2XKO](https://www.riotgames.com/en/games/league-of-legends) (prev. Project L).

Research Programmer Intern and Tools Programmer Intern, Epic Games - Jun 2020-Jan 2021 - 40 h/wk - Remote - [unrealengine.com](https://www.unrealengine.com)

Created deep reinforcement learning samples, a machine learning plugin, and virtual production tools in Unreal Engine.

Google Engineering Practicum Intern, Google Cloud Platform - Jun 2019-Sep 2019 - 40 h/wk - Seattle, WA - github.com/knative-portability

Developed full-stack [open-source applications](https://github.com/knative-portability) as proof of portability for [Knative](https://knative.dev), a platform for serverless containerized workloads.

Selected Publications:

Escalation Risks from Language Models in Military and Diplomatic Decision-Making - Oct 2023-Jan 2024 - [Paper](#), [GitHub](#)

Co-first author. Evaluating the national security risks from autonomous language model decision-makers in escalating international conflicts. Accepted to ACM FAccT 2024, MASEC NeurIPS 2023 workshop (spotlight).

Welfare Diplomacy: Benchmarking Language Model Cooperation - Jun 2023-Sep 2023 - [Paper](#), [GitHub](#)

First author. Multi-agent LLM evaluations in a novel general-sum variant of Diplomacy that better incentivizes and measures cooperation. Accepted to the SoLaR NeurIPS 2023 workshop.

SuperHF: Supervised Iterative Learning from Human Feedback - Jan 2023-Sep 2023 - [Paper](#), [GitHub](#)

First author. Alternative to RLHF post-training via supervised learning without RL. Accepted to the SoLaR NeurIPS 2023 workshop.

Societal Adaptation to Advanced AI - Jan 2024-May 2024 - [Paper](#), [Thread](#)

Co-first author. AI strategy and policy analysis discussing how society might adapt to future risks from advanced AI systems.

Skills:

Artificial Intelligence - software.gabrielmukobi.com/ai

AI security, NLP, AI governance, evaluations, ML, DL, foundation models, research mentorship. Languages: Python.

Software Engineering - software.gabrielmukobi.com

Product management, documentation, testing, bug reporting, code review, CS, VCS, [GitHub](#), [GitLab](#). Languages: Python, C++, C#.

Web Development - software.gabrielmukobi.com/web

Full-stack, web design, cloud computing, databases, Docker containerization. Languages: JavaScript, Node.js, Python, HTML.

Education:

University of California, Berkeley - Ph.D. Computer Science - Aug 2024-Leave

Ph.D. student on leave researching machine learning security. Advised by [Jacob Steinhardt](#) and [Dawn Song](#).

Stanford University - M.S. Computer Science - Sep 2023-Mar 2024, B.S. Computer Science - Sep 2018-Dec 2023 - Cum GPA: 4.01

Coursework and extracurriculars in AI/ML, Graphics, Computer Systems, Algorithms, and Theory.